Step 0: Pre-Analysis Reflexive Declaration

Home Grammar: My default lens is the mainstream biomedical/public-health framework. I trust clinical trials, regulatory processes, and official health guidance. I tend to see scientific and institutional factors as primary causes (e.g. virus biology, vaccine immunology) and weigh evidence like peer-reviewed data most heavily.

Hardest Grammar: Operating a strongly anti-vaccination or conspiratorial grammar will be hardest. It requires downplaying mainstream science and trusting fringe interpretations of signals (e.g. anecdotal adverse events or alleged cover-ups). This framework is far from my default trust in institutions.

Potential Blind Spots: From my position, I might underappreciate concerns about rare side effects, individual rights issues, or legitimate questions about equity and consent. I may also downplay how industry motives or political factors can influence the public discourse.

Temporal Orientation: I am oriented mainly towards *causal explanation* (looking backward): I focus on how vaccine development and rollout caused epidemiological outcomes. I also consider forward-looking policy implications (mitigation, boosters). I am less attuned to purely ideological or meaning-making interpretations that are not causally driven.

Section 1: Observation Field Declaration

Phenomenon: We analyze the COVID-19 vaccination discourse (2020–2024), including vaccine development, authorization, distribution, uptake, and the rise of vaccine hesitancy/anti-vaccination movements. Geographically, this spans much of the world, with emphasis on high- and middle-income countries heavily involved in early vaccine rollouts (e.g. US, EU, UK, Israel) and accounts for global trends.

Interpretive Divergence: This topic produces sharp divergence because it intertwines scientific data (vaccine efficacy/safety), political decisions (mandates, approvals), and public values (trust, autonomy). The same inputs – e.g. a VAERS report or a trial result – are understood very differently by different communities. Strong emotions (fear of disease vs. fear of side-effects) and ideological factors (trust in government, libertarian views) further polarize interpretation. (For example, WHO director-general António Guterres warned of an “infodemic” as vaccine misinformation exploded online

.)

Signal Contestation Level: The domain is **highly contested**. Many core signals (e.g. adverse event data, excess deaths) are interpreted differently across groups. Some signals (like vaccine efficacy in trials) are relatively uncontested, but others (like the meaning of VAERS reports or trends in all-cause mortality) are hotly disputed.

Major Interpretive Communities: Key communities include:

- **Mainstream Public Health/Science:** Government health agencies, clinical scientists, epidemiologists, and much of the medical media. They prioritize clinical trial data, population-level outcomes, and risk–benefit analyses, generally concluding “vaccines save lives.”
- **Vaccine-Skeptical/Risk-Oriented:** This includes vaccine-hesitant advocacy groups, some disaffected medical voices, and sections of social media. They emphasize reports of adverse

effects, regulatory “shortcuts,” and worst-case scenarios, often concluding “the risks of vaccines may outweigh the benefits.”

- **Libertarian/Political Freedom Advocates:** Policymakers and commentators focused on individual rights. They may accept scientific efficacy in principle but oppose mandates or passports. They treat coercion or regulation as the primary lens (e.g. “mandates cause distrust”). This community spans from moderate to radical libertarian, often skeptical of institutions due to principle rather than evidence alone.

Each of these communities shares some evidence with the others but applies different filters and priorities. The prismatic analysis will make these filters explicit.

Section 2: Signal Set

Below are the **core signals** (key empirical inputs), each with source(s), type, and an ambiguity note.

- **Clinical Trial Efficacy (Pfizer/BioNTech):** Phase III trial reported ~95% efficacy (95% CI 90.3–97.6) after 2 doses

, with ~85% efficacy against severe disease , with full approval for Pfizer in Aug 2021 . Combined protection against severe disease fell to ~64% at ~121 days . FDA/CDC concluded myocarditis is causally linked to mRNA vaccines . It also found government measures and vaccination correlated with *lower* excess mortality (expecting $32B in 2022); Moderna expected ~$18.4B in 2021 , later converting to a stricter “vaccine pass” (vaccination only) in Jan 2022). Simultaneously, trust in health agencies fell: a Jan 2025 KFF poll showed trust in the CDC had fallen to 61% (down from 66% in June 2023) . In the US, rural counties lagged urban (58.5% vs 75.4% first-dose coverage by Jan 2022). A French analysis found that living in poorer, higher-immigrant, or low-trust districts reduced vaccination probability by ~30%

- . *Type:* Demographic analyses. *Ambiguity:* Underscores equity issues; some see these as evidence of access and trust gaps needing policy, while contrarians sometimes misinterpret them (e.g. attributing low minority uptake to “them being anti-vax”), which mainstream experts caution against.

Counter-Signals (Complicating Evidence):

- *Lives Saved:* Studies estimate vaccines prevented *millions* of deaths. For example, COVID vaccines have been credited with saving ~19 million lives globally in their first year
- . If a grammar expects “widespread harm,” this is a tension.

Contested Signals: Certain inputs themselves are disputed between grammars:

- **VAERS Reports:** *According to public-health grammar,* VAERS is an early warning system; any individual report is not proof (CDC: “A report to VAERS does not mean the vaccine caused the event”
-). *Skeptic grammar* might claim any post-vaccine-year excess was caused by vaccines. Thus “excess mortality” is a *contested signal*.
- **Regulatory Speed:** Fast EUAs mean “urgency” to one grammar, but “insufficient testing” to another. The *signal itself* (rapid approval) is contested.

Each of the above will be explicitly flagged when it appears in processing.

Section 3: Grammar Specifications

We will construct three grammars representing major interpretive frameworks. Each is defined by its required and recommended parameters.

Grammar A: *Biomedical/Public-Health Model*

- **Primary Analytical Variable:** *Pathogen/disease risk and immunity.* This grammar views SARS-CoV-2's biology and population immunity as causally fundamental. Vaccine immunogenicity and virus transmission dynamics are primary drivers.
- **Signal Selection Rules:** Admits peer-reviewed clinical and epidemiological data, official stats (CDC, WHO, clinical trials, surveillance). Deems regulatory and large-cohort studies admissible signals. Excludes signals lacking formal scientific validation (e.g. isolated anecdotes, unexplained correlations without statistical basis).
- **Weighting Priority:** Prioritizes high-quality evidence: randomized trials > large observational studies > case series > expert consensus. Signals directly tied to disease outcomes or vaccine efficacy get top weight. Adverse-event frequencies are weighed in context of background rates. Economic/motivational signals are secondary to health outcomes.
- **Causal Chain Type:** Mechanistic-temporal. It builds a linear causal chain: virus ➔ (interventions like vaccines) ➔ population health outcomes. Explanation focuses on how vaccines interrupt infection chains or mitigate disease, often quantified by models (e.g. reduction in R_0, prevented hospitalizations).
- **Admissible Closure:** Conclusions must be consistent with medical-scientific plausibility. The range includes "vaccines significantly reduce disease burden; recommended for all eligible" to nuanced "vaccine program reduced overall mortality but waning/variants require boosters". It cannot conclude vaccines "created the virus" or that vaccination policies are deliberate tyranny.

Recommended parameters:

- **Excluded Signals:** Ignores unverified personal testimonies as evidence; spiritual or ideological arguments (e.g. claims that vaccines go against nature) are irrelevant. Suspicion-based signals (like "they withheld cure") are excluded.
- **Characteristic Vocabulary:** "Efficacy," "clinical trial," "statistical significance," "immunogenicity," "endemic control," "safety profile," "public health impact," etc. Language is technical and cautiously optimistic (e.g. "highly effective," "rare adverse events").
- **Temporal Orientation:** Primarily *backward-causal*, tracing how interventions changed infection/mortality curves. Also *forward-looking* in prescribing next steps (e.g. booster campaigns). Tends to treat guidance changes as updates improving accuracy.
- **Temporal Formation:** Formed during early vaccine trials (~2020) and solidified by late 2021 (Omicron). Subsequent evidence is incorporated (some as confirmation of predicted waning, some as refinements). Challenging data (e.g. side effects) are noted but usually framed as manageable exceptions to the overall benefit.

Source Tradition: This grammar draws on public health and medical research (CDC/WHO briefings, epidemiology journals, textbooks of vaccinology). Its arguments resemble those in medical guidelines and policy papers.

Grammar Boundaries: It would transform into Grammar B/C if the foundational variable shifted. For example, if one began treating *vaccine components or corporate motives* as "causally fundamental" instead of pathogen biology, it would tilt into skeptic territory. If *individual liberty* became paramount, it would become a civil-libertarian grammar.

Grammar B: *Safety-Skeptic/Anti-Vax Model*

- **Primary Analytical Variable:** *Risk from vaccine or authorities.* This grammar treats potential vaccine harms (or perceived overreach) as the fundamental causal factor. It often posits hidden or long-term risks from medical interventions as driving outcomes.
- **Signal Selection Rules:** Admits signals that imply risk or mismanagement: passive surveillance reports (VAERS), media anecdotes of adverse reactions, anomalies in data (e.g. clusters of non-COVID death), and testimonies from alleged whistleblowers. It is skeptical of official trial data unless independently reproduced. Excludes signals that downplay harm (e.g. average-case modeling without showing individual risk) or those seen as conflicts of interest (industry-funded studies may be dismissed).
- **Weighting Priority:** Ranks signals that indicate danger or conflict highest. A small number of serious adverse events (e.g. reports of myocarditis or deaths after vaccination) are weighted heavily (often as representative, not probabilistic). Testimonials or leaked docs may rank above aggregated statistics. Profit or political signals (e.g. company revenues, lobbying) are given significant weight as motives.
- **Causal Chain Type:** Often *disjunctive or conspiratorial*. Causation may be seen as multi-layered: e.g. “Big Pharma profits -> regulatory shortcuts -> mass vaccination -> widespread harm.” It permits looping or feedback: a single adverse event can be seen as affecting trust and uptake. Explanation may invoke intentional or negligent causal links rather than purely biological mechanisms.
- **Admissible Closure:** Allowed conclusions range broadly but generally revolve around “vaccination is net harmful or unnecessary” or “the authorities lied.” It can conclude vaccines are dangerous, mandate-driven coercion is illegitimate, etc. It does **not** accept closures that assume general efficacy if signals strongly suggest danger (e.g. “the vaccine saved millions” is outside its closure).

Recommended parameters:

- **Excluded Signals:** Disregards evidence of population-level benefits as “the average hides the truth.” Disbelief in randomized trials (seen as manipulated). Excludes statistics standardized away from raw counts (prefers raw VAERS numbers).
- **Characteristic Vocabulary:** Terms like “adverse reaction,” “injury,” “suppress,” “cover-up,” “toxic,” “mandate,” “billions in profits,” “industrial collusion,” “breakthrough,” “genocide” (in extreme cases), etc. Rhetoric is often alarmist or emotive.
- **Temporal Orientation:** Tends to look *backward/historical*, seeking “hidden causes.” There’s focus on what was *done* wrong (e.g. rushed EUA) or what harm emerged. It can be prescriptive in warning “never again” (pseudo-future, “Never again should…“). This grammar forms early closure and then treats later data as vindication or, if contradictory, as “cover-up.”
- **Temporal Formation:** Likely formed from early 2020 skepticism (possibly rooted in pre-COVID anti-vax sentiment), crystallizing by 2021 as reports of side-effects emerged. New evidence (e.g. studies showing waning, or Omicron data) are often absorbed as confirmation (“see, vaccines don’t work anyway!”) or reinterpretation.

Source Tradition: Draws on alternative-media reporting, independent bloggers, and distrustful investigative journals. Influences include past vaccine scares (e.g. MMR-autism controversy), anti-pharma activist publications, and certain social media communities.

Grammar Boundaries: To become Grammar A, this view would have to start prioritizing large-scale clinical evidence over anecdotes (e.g. if it accepted trial methodology as valid). To become Grammar C, it would need to shift from “vaccine danger” to a principled “government overreach” focus; in practice, a skeptical standpoint with stronger emphasis on rights.

Grammar C: *Libertarian/Autonomy Model*

- **Primary Analytical Variable:** *Individual freedom vs. state power.* Causally fundamental is the relationship between citizens and authorities. Actions are interpreted through the prism of coercion or consent, not primarily by biomedical metrics.
- **Signal Selection Rules:** Admits signals of mandates, restrictions, or government/institution overreach. For example: dates of imposing vaccine passports, fines for noncompliance, funding or legislation behind mandates. Also notes political statements about "common good" or "herd immunity" as signals. Technical efficacy or safety data is secondary or filtered: accepted only insofar as relevant to policy rationales. Excludes pure health outcomes that do not touch on autonomy (e.g. raw mortality curves) unless directly linked to mandates.
- **Weighting Priority:** Highest weight to any signal about compulsion or choice restriction. Individual cases of rights infringement (e.g. healthcare worker firing over vax status) are weighed heavily. If vaccine/vaccination is voluntary, signals of success/efficacy are less decisive. Economic or profit signals are considered only as they motivate coercion.
- **Causal Chain Type:** *Normative-social.* The chain explains outcomes as consequences of policy on public trust and behavior. Example chain: "Mandates and censorship by authorities → Erosion of trust and increased resistance → Lower overall compliance." Explains phenomena by values (freedom, distrust) rather than virology. Causality can be circular (e.g. policy leads to backlash which affects subsequent policy).
- **Admissible Closure:** Conclusions focus on rights and consent: e.g. "No mandate should be imposed regardless of efficacy" or "Without personal consent, any vaccination program is illegitimate." It's permissible to conclude that even a very effective vaccine shouldn't be forced. It cannot conclude "vaccines are inherently a government conspiracy" unless government missteps are explicit; it **must** preserve a thread of individual choice in closure.

Recommended parameters:

- **Excluded Signals:** Ignores broad epidemiological benefits (e.g. "many lives saved") as secondary to the tyranny-issue. Technical safety statistics can be admitted only to the extent they bolster the claim "government forced something dangerous." Scientific terminology is used only rhetorically (e.g. quoting CDC to show how they changed mind).
- **Characteristic Vocabulary:** "Freedom," "liberty," "consent," "bodily autonomy," "infringement," "mandate," "passports," "Big Brother," "government overreach," "medical freedom," etc. Emphasizes legal-constitutional terms.
- **Temporal Orientation:** Mixed lateral/forward. Often frames issues as present ethical dilemmas ("New mandates will erode trust"). Historical analogies (past government abuses) used hermeneutically. It's partly *forward* in arguing what society should do (e.g. "Resist mandates now to preserve freedom"). Formation occurs around major policy changes (e.g. mid-2021 passport debates), and then views new evidence as either vindicating or challenging rights claims.
- **Temporal Formation:** This grammar crystallized around the time when governments began mandates/passports (~mid-2021). Prior "formation" includes earlier libertarian views on healthcare. Evidence since then is interpreted through the autonomy lens: e.g. if a mandate is lifted, it's seen as cause for vindication, not as evidence it was unnecessary (because the grammar focuses on "it was unfair anyway").

Source Tradition: Draws on libertarian political philosophy and legal arguments, as well as civil-rights activist literature. Influential texts include charters of rights, civil liberties organizations' statements, and some conservative/libertarian think-tank publications.

Grammar Boundaries: It would shift to Grammar A if it started treating outcomes (cases/deaths) as more important than autonomy, or to Grammar B if it began to attribute issues primarily to the vaccines themselves (rather than to policy). A change from "freedom" to "safety" as core would cause a flip.

Section 4: Signal Processing

Below, each grammar processes the core signals through its lens. Within each, contested signals and counterpoints are noted in context. Each grammar's *closure* is presented at the end of its section.

Grammar A (Biomedical/Public Health) Processing

This grammar treats **vaccine efficacy data** as validation of the vaccine strategy. The ~95% trial efficacy for Pfizer

and 94.1% for Moderna are viewed as extremely strong signals of success. It gives these figures highest weight: "Two-dose mRNA regimens prevent roughly 9 out of 10 cases in trial settings." AstraZeneca's 70.4% efficacy is seen as lower but still beneficial, especially given variant context. J&J's 52.9% moderate efficacy is noted as weaker but still valuable, particularly with its 85% severe-case efficacy

as evidence it prevents the worst outcomes. In summary, **all trial signals** lead the grammar to conclude "vaccines are highly effective in preventing COVID, especially severe illness."

Regulatory/EUA signals: The rapid EUAs (Dec 2020 for Pfizer/Moderna

) are seen as necessary responses to a public health emergency. Expanding to younger ages (12–15 in May 2021) is interpreted as evidence of safety confidence. The August 2021 FDA full approval (91% efficacy in updated data

) reinforces trust: "FDA found high efficacy again and transitioned to full approval." Any skepticism in the EUA timeline is downplayed: the grammar notes that all EUAs were based on "substantial evidence" from large trials, so their speed is a feature, not a bug. It flags no "problem" with emergency authorization per se.

Waning and Variants: Real-world effectiveness and waning are integrated as refinements, not contradictions. The Swedish cohort study

(showing >90% initial protection fading to negligible by ~7 months) is accepted and weighted: "Vaccines protect strongly early but immunity declines over months." The grammar notes it still found some protection against severe disease (64% at ~4 months). It uses this to motivate boosters. Likewise, Omicron performance is integrated: the grammar acknowledges that Omicron largely escapes vaccine protection against infection (only ~26–35% at 100 days post-booster

), but likely infers that **severe disease protection is still non-zero** (the grammar would point out that VE figures were for infection, not hospitalizations). Thus, the grammar would incorporate these signals by adjusting the model: "Boosters needed to maintain protection, and high-transmission variants require additional measures, but fundamentally vaccines still mitigate the worst outcomes."

Adverse Events (Safety) signals: This grammar takes these signals seriously but in context. It recognizes CDC's classification of myocarditis as causally linked to mRNA vaccines

, and TTS with adenovirus vaccines leading ACIP to favor mRNA. However, it weighs the *frequency and severity* of those events: e.g. myocarditis occurs mostly in young men and is generally mild/recoverable, so the grammar notes "risk is rare and outweighed by vaccine benefit." It similarly notes Guillain–Barré risk after J&J (reported around 1 per 100k doses, though CDC status is monitoring), but places it in perspective of COVID-caused neurologic damage. The *counter-signal* "thousands of myocarditis reports in VAERS" is addressed by citing CDC/VAERS context

: “A VAERS report does not prove causation, and many reports are unverified.” This grammar will emphasize active surveillance (e.g. VSD studies) showing only a few extra cases per 100k, not catastrophic rates. Any counterpoint within safety is acknowledged: **“Some rare harms exist,”** but by its rules the conclusion remains “vaccination’s benefits far exceed these small risks.” No closure will assert “vaccines are dangerous” because that violates its admissible closure.

Excess Mortality: The grammar accepts WHO’s estimate of 14.9M excess deaths in 2020–21

as evidence of the pandemic’s toll. Since 2022 still had high excess deaths in many places

, the grammar interprets this as an *indirect pandemic effect*. It cites [47†L649-L653]: “Excess deaths were mainly non-COVID and attributed to pandemic disruptions” (e.g. delayed care). The signal “excess mortality persisted even post-rollout” is thus understood as partly due to earlier infection waves, healthcare backlog, and new variants, *not* as a surprise that undermines vaccination. In fact, the grammar highlights [47†L649-L653]: “Government control and vaccination in 2021-22 were associated with *reduced* excess mortality,” so vaccines are seen as mitigating even indirect deaths. A *contested counter-signal* here might be the anti-vax claim “excess deaths prove vaccine harm” – the grammar counters by noting epidemiologists attribute excess 2022 to indirect causes, not vaccines.

Pharma Profits: The grammar is aware of the $36.8B Pfizer sales

and $18.4B Moderna orders. These are noted as evidence of high demand and return on investment. It does not consider profit as inherently bad; rather, it may see it as a return that funds future R&D. The grammar might cite [66†L364-L367] as fact (“Pfizer expects $32–37B from vaccines”) but frames it positively (“high manufacturing capability”

) or neutrally: “Yes, vaccines generated major revenue, reflecting unprecedented global need.” The safety-skeptic *counterpoint* that “Big Pharma is the real target” is discounted: business success is not taken as evidence of harm to individuals. This grammar’s weighting still is on health outcomes, not money.

Government Mandates/Passports: Mandatory vaccination policies are interpreted as public health measures. For example, France’s sanitary/vaccine pass (Aug 2021, Jan 2022)

is seen as an aggressive but effective campaign: “France used passes to boost uptake; this signals the state prioritized controlling spread.” The grammar notes any increase in vaccination from those measures as success. It may mention controversy but will view opposition as “marginal backlash.” If encountering a counterpoint (e.g. some claims it had no effect), it will demand data; lacking that, it concludes mandates were justified by the large risk. It might acknowledge rollout pains (e.g. “some people were upset”) but closure remains that “mandates, while onerous, served the greater public health goal.”

Public Messaging Changes: Frequent guidance changes (like mask advice, boosters) are interpreted as normal science updating. The grammar will say e.g. “WHO/CDC adjusted guidelines as new data (e.g. waning) emerged.” If a counter-signal is that “constant change means they didn’t know what they were doing,” Grammar A would reframe it as scientific learning (“guidance evolves with evidence”). Thus this grammar assimilates the timeline as a rational progression.

Social Media Moderation: Actions like Meta’s removals of false claims

are seen as necessary to protect the public. The grammar would note [78†L309-L317]: “Facebook expanded its misinformation removals about vaccines” – and interpret that as an alignment with WHO advice (it even consulted WHO

). Any critique of “censorship” is dismissed: “misinfo has real harm,” so removing it is not problematic. In summary, Grammar A trusts the platforms’ mitigation efforts as justified.

Public Opinion Polls: Strong support (~90%) for vaccine access and trust in science

is interpreted as baseline consensus: “Most people get it.” The grammar notes [88†L79-L87]: “A vast majority of Americans want vaccines to be accessible.” It will also note [80†L320-L328]: trust in CDC/FDA declined to ~61% by 2025. This poses a counterpoint *within the grammar*: “Why is trust falling?” It attributes it to political attacks (e.g. partisan divide

) or misinformation, not to scientific faults. The grammar concedes some credibility to the signal (trust decline) but sees it as a communication problem rather than invalidating the vaccines. Thus, it may note “even if some distrust exists, overall acceptance remains high.”

Vaccination Disparities: The grammar accepts disparities as concerning signals. For example, [83†L229-L237] (Blacks 13% less likely to be vaccinated) is a signal that inequities exist. It sees this as a call for targeted outreach, not evidence of failure of the vaccines themselves. Rural–urban gap [86†L61-L64] similarly indicates access issues. The grammar’s closure doesn’t change (vaccines are still effective), but it acknowledges these disparities as “social determinants” issues. It would incorporate them by suggesting policy remedies.

Closure (Grammar A): Vaccines are *safe and effective public-health tools*. All evidence from trials and real-world data shows high protection (especially against severe disease), and while immunity wanes (necessitating boosters), overall benefit far outweighs small risks. Government programs and high coverage saved lives (WHO’s excess-death analysis

). Differences in uptake are regrettable but addressable by outreach. The recent trust decline is an urgent issue but does not negate the scientific consensus. This grammar concludes that widespread vaccination should continue (possibly with adaptations for variants), and that mandates, while tough, were defensible for the common good.

Note: This closure is *not* a verdict that absolutely “all vaccines are perfect” or that “no one is harmed.” It is the *product of Grammar A’s architecture* focusing on overall epidemiological logic, not a commentary on individual beliefs. It emphasizes that this is the conclusion *given its rules of evidence*, not “the final truth” in other frameworks’ terms.

Grammar B (Safety-Skeptic) Processing

Grammar B starts by treating **vaccine trial efficacy** data with suspicion. The ~95% and 94% figures

are noted but bracketed by caution: “These were under ideal trial conditions on selected participants.” It looks for “nuance”: Pfizer’s 95% came from 2-month follow-up, and even FDA later said “91% effective” in real use, implying the vaccine *already lost some efficacy over time*. It highlights such *updated values* as evidence of decreasing effect. It also notes AstraZeneca’s varied efficacy and Johnson & Johnson’s modest figures

to argue “no vaccine is uniformly miraculous.” Any claim of broad success is muted: it might concede “vaccines can reduce disease, but efficacy is not near-100% in practice.”

The **EUA timeline** is interpreted critically: e.g. Pfizer’s EUA on Dec 11, 2020

is framed as “only emergency authorization, not full approval.” It stresses that by Aug 2021, after billions of doses, Pfizer got full approval only after internal data reanalysis (“Before that they had no idea what it did after 6 months”). If regulated slower or more transparently, we’d trust more. Similarly, Moderna’s Dec 18,

2020 EUA is noted as “second vaccine rushed.” Grammar B might highlight any hesitations or pauses: e.g. J&J’s Feb 2021 EUA came with a 10-day pause for TTS

, suggesting “even regulators needed a timeout.” Overall, EUA signals are spun as “the process was politicized and fast-tracked at the expense of caution.”

Waning/Variants: The Swedish waning study

is embraced as a key signal: “Yep, immunity drops to zero after months for Pfizer.” Grammar B says “See, vaccines *failed to protect long-term*.” It weights that more than initial benefits. It also notes the severe-disease efficacy fell to only 64% at ~4 months, which is cast as “partial failure.” The Omicron data

is similarly seized: “Even boosted, only ~26–35% effective against infection after a few months!” The grammar would likely *overemphasize* infection prevention rates as if preventing any infection is the goal, arguing “with such low VE, the whole premise of vaccination is shaky.” It may underweight (or ignore) the fact that these are VE against infection, not death. A grammars-level counter-signal (“primary outcome is death/hospitalization”) is seen as dismissible.
Importantly, this grammar includes the *counterpoint from within*: it admits **some** protection exists initially, but frames it as transient or limited. It may concede “it’s better than nothing,” but often its tilt is “not nearly as effective as claimed.”

Safety/Adverse Events: This grammar places extreme weight on adverse-event signals. It treats raw VAERS totals as meaningful: “Thousands of deaths and injuries have been reported.” It likely cites the number of reports (even though unverified) as a red flag. The official CDC caveat that VAERS ≠ causation

is downplayed or called “bureaucratic spin.” Instead, each reported myocarditis or TTS case is treated as evidence. The confirmed signals (myocarditis, TTS

) are underscored: “They admit these bad outcomes are happening.” Grammar B may create chains like: “mRNA vaccine → myocarditis in youth; adenovirus vaccine → deadly clots.” Any mention that these events are “rare” in clinical terms is seen as minimization. If at all, it would say “even one avoidable death or injury is too many!”

Counter-signals about safety (like many safe doses given, no spike in known conditions beyond these signals) are treated skeptically. For example, the grammar might see CDC’s slide stating vaccines saved lives

and question it: “How can you prove lives saved? They just did models.” If [39] says “myocarditis is causally linked,” it concludes “confirmed danger.” It might dismiss [38] entirely. The grammar also often cites VAERS-derived analyses (e.g. “We observed far more heart inflammations than expected baseline”), giving these quasi-studies great weight.

Excess Mortality: Given the heavy emphasis on risk, excess deaths are interpreted as red flags. Grammar B sees the 14.9M WHO excess

and may allege “this is partly vaccine death.” It *contests* the mainstream explanation: e.g. the C-MOR study said “excess 2022 was non-COVID”

, but Grammar B says, “What’s more likely? It’s COVID or vaccines causing these mysterious deaths.” It does not accept [47]’s causal analysis at face value. Instead, it highlights any negative mortality trends after vaccination campaigns as potential “cover-ups.”

Pharma Profits: The huge revenues

are taken as evidence of motive. Grammar B will argue “Companies stood to gain billions, of course they’d push vaccines aggressively.” It might link [66] and [72] to a causal story: “Billions in profits → bribery/lobbying → fast approvals → mass vaccination push.” It would consider profit signals core, not ancillary. Possibly references “Billions of dollars spent on lobbying for mandates.” It may call these profits “blood money.” (The inflection is strongly negative.)

Mandates/Passports: Signals of coercion are primary here. France’s sanitary/vaccine passes

are cited as examples of “government forcing medical intervention.” This grammar treats mandates themselves as *evidence of wrongdoing*: e.g. “If it’s so great, why make it compulsory? That proves people wouldn’t take it otherwise.” Any evidence that mandates increased uptake is reinterpreted as “herd mentality” or “fear.” It might downplay the logic of protecting others. The *counter-signal* of mandates being lifted (in some places) is seen as victory: “Authorities finally admit it was overreach.” It does not credit mandates for any health benefit.

Public Messaging/Trust: This grammar pays attention to shifts as signs of inconsistency: e.g. changes in CDC guidance suggest “they don’t know what they’re doing.” The KFF poll showing trust drop

is seen as proof that the public is waking up. It may cite [80†L320-L328] to say “Even trust fell to ~60%; people are skeptical.” Lower trust among Republicans is not “just politics” but seen as correct skepticism. The grammar cites any slip or retraction (like correction of disallowed claims – or the earlier WHO mention of infodemic) as validation of distrust.

Social Media Moderation: Sites’ anti-misinformation rules

are interpreted as *suppression of truth*. Grammar B will say, “Look, Facebook is removing *accurate* claims like ‘vaccines cause autism’!” It will likely quote the very bullet “Vaccines are toxic, dangerous or cause autism” as an example of what they forbade

, implying that legitimate concerns are being silenced. The label “remove false claims” is inverted to “this label *is* false.” In short, the grammar sees content moderation as censorious and further evidence of a cover-up.

Public Opinion: The high acceptance polls

are received with suspicion: “sure, people support access *in general*, but they don’t know the truth.” Grammar B tends to think surveys are rigged or reflect social desirability. It may focus on the ~17–26% minority skipping vaccines or the 78% who worry about political interference

as evidence that “many are wary.” For example, it will seize [88†L99-L103]: “78% say politicians are messing with vaccines – that’s a lot!” It interprets that not as a footnote but as core: “Majority suspect foul play.”

Vaccination Disparities: This grammar might use disparities to claim “they are targeting certain groups for harm.” E.g. lower minority rates

can be spun as evidence those communities were “sacrificed” or lied to. Or it might absurdly say “Why are white people getting higher rates? Because they comply blindly.” It’s selective: if a signal shows a minority was targeted (e.g. giving them fewer doses), it’s proof of inequity. If it shows minority uptake was lower, it’s proof of distrust. In any case, disparities are taken to underscore either nefarious policy or awakened skepticism among disadvantaged groups.

Internal Counterpoint: Within Grammar B’s logic, a possible *counter-signal* is that *many people did get vaccinated voluntarily*. The grammar must confront the fact that some trust or accepted vaccines. It might handle this by splitting: “OK, most did at first, but only under propaganda or fear.” It might call early high

uptake “mass panic reaction” and say now people see through it. Thus, voluntary uptake is not taken as a refutation but as a temporary effect.

Closure (Grammar B): Vaccination campaigns have been a case of “unknown, likely harmful medicine” pushed on the public. The processing above leads to the closure that **we cannot trust that COVID-19 vaccines are safe or effective**, and that the official narrative is driven by profit and control. It concludes that ongoing vaccination (especially mandates) is unjustified and possibly dangerous. It might allow for a very narrow place-holder closure like “if anything, we *should* stop/withdraw the vaccines and investigate safety further.” But chiefly, closure asserts that evidence for harm > evidence for benefit, and skepticism should prevail.

Note: This grammar’s closure is a product of its selectivity and causal framing. It does not claim an omniscient verdict on the virus itself, only that the vaccines (as social phenomenon) are net negative under its rules.

Grammar C (Libertarian/Autonomy) Processing

Grammar C focuses on **authority and choice** as lenses, rather than medical signals. It treats many health-related signals as secondary or as context for power issues.

- **Efficacy Data:** The trial numbers

themselves are not immediately central. This grammar might gloss over the figures, noting simply that “the vaccines were said to be highly effective, but that isn’t the issue – being forced matters more.” It allows efficacy signals but reframes them: e.g. “Yes, Pfizer claimed ~95%, but none of this should be mandated.” If confronted with waning/variant data and clots. If it’s not 100% safe, how can they *mandate* it on everyone? That’s coercion with risk.” It uses the mere existence of acknowledged harms to fuel “damned if you do, damned if you don’t” logic: if vaccines are safe, we shouldn’t need mandates; if not safe, mandates are intolerable. VAERS reports – if many died anyway, how do we know vaccines helped?” Or “Look, year 2022 still had excess mortality are seen positively (good that people want availability). But the grammar focuses on the *nuance* that many also worry about politics in vaccines: [88†L99-L103] notes 78% think interference is happening. It will cite this as evidence of skepticism. The grammar might say: “Even among the 90% supporting vaccines in general, a majority believes the process is politicized.” The trust-institutions decline

- shows “states with mandates still failed their rural citizens.” It may claim mandates hurt rural communities who have little choice. It might argue the converse: “Why force shots when rural people still lag? Perhaps policy is misaligned with people’s values.” It interprets disparities as symptoms of trust-issues, not as triggers for more compulsion.

Internal Counterpoint: The biggest counterpoint to this grammar is that *most people did take the vaccine voluntarily*, which complicates an autonomy-only stance. The grammar handles this by saying the majority weren’t *strictly forced* (no one had a gun to their head); thus high uptake reflects trust of science but doesn’t negate the right of dissent. If pressed, it may concede that most acted “prudently,” but insist dissent should still be protected.

Closure (Grammar C): No analysis of data will outweigh the primacy of individual choice. The processing leads to the conclusion that **vaccination should be a free personal choice, not mandated by government or corporations**. Even with high efficacy or saving lives, if it infringes on autonomy, it is too much. The grammar likely calls for protecting bodily autonomy in law (e.g. forbidding mandates). Any benefits of vaccines are secondary to the citizen’s right to accept or refuse.

Note: This closure stems from valuing freedom as the core variable. It is *not* a statement about efficacy or safety per se; it's a constitutional/political stance produced by this framework.

Section 5: Closure Comparison

Closure Content:

- *Grammar A (Biomedical)* concluded: *"Vaccines are demonstrably effective and, despite some rare risks, have saved many lives; continued public vaccination programs (with booster updates) are justified."*
- *Grammar B (Safety-Skeptic)* concluded: *"The risks and unknowns of COVID vaccines outweigh claimed benefits; the official narrative is suspect; vaccination campaigns (especially mandatory ones) should be halted."*
- *Grammar C (Libertarian)* concluded: *"Vaccination must remain an individual choice; any coercion (mandates/passports) is illegitimate, regardless of efficacy claims."*

Each grammar focuses on different aspects of "what to do about vaccines," so the exact statements differ substantially. Grammar A's closure is basically endorsing vaccination programs; Grammar B's is to mistrust/discontinue them; Grammar C's is to protect choice above all.

Signal Divergence:
The signals that generated the greatest interpretive splits include:

- *VAERS Adverse Reports:* A primary driver of divergence. Grammar A largely disregards unsubstantiated VAERS signals
- , B often blames them on vaccination side-effects (or at least casts doubt on official explanation), and C uses any residual excess as evidence that policy hasn't "solved" the crisis, emphasizing the failure of collective measures.
- *Pharma Profits:* Grammar A notes profits but doesn't alter its health-based view; Grammar B treats large revenue as a key signal of malintent; Grammar C uses profits as evidence of an industry-driven system but focuses on how that underwrites coercive policy.
- *Mandates/Policies:* Grammar B and C both treat mandates as deeply negative, whereas Grammar A treated them as necessary. The presence or absence of strict policies largely determines whether B/C feel vindicated.
- *Waning/Variants:* All grammars incorporate waning, but A sees it as a technical issue to fix with boosters (continuing vaccination effort), B sees it as evidence vaccines aren't worth the trouble, and C sees it as further reason not to impose recurring vaccinations.

Exclusion Patterns:

- Grammar A *excluded* grassroots opinion and ideological critiques (like "it violates rights") – it sees them as outside valid evidence. It similarly excluded raw VAERS causation.
- Grammar B *excluded* large-scale mortality and efficacy data as "misleading" (it used lower efficacy data when convenient, but generally stayed critical). It excluded rationale about public benefit.
- Grammar C *excluded* population-level benefit analysis as irrelevant; it prioritized personal and legal language. It also excluded personal anecdotes of adverse events unless they pointed to forced injection.

So, for instance, Grammar A “could see” private concerns as noise, but B/C would certainly highlight them. Grammar B could not accept the official narrative of “deaths averted,” whereas A/C would accept that view (A analytically, C possibly rhetorically). Each grammar left some other-grammar signals on the table.

Causal Architecture Differences:

- Grammar A’s chain is linear (virus→vaccine→reduced cases/deaths). It attributes population outcomes mainly to biomedical actions.
- Grammar B’s chain is often circular or conspiratorial (pharma->media->people->sick people; or vaccine->unknown harms->cover-up->public mistrust). It allows hidden intermediaries (e.g. profit motives) and feedback (media shaping risk perception). Causal statements often emphasize intentionality or negligence.
- Grammar C’s chain is normative/social (policy→trust/behavior). It sees causes in terms of power relations. It might not even frame traditional “cause” – instead “If a mandate is enforced, then people feel violated.” It does not analyze biological causation at all; “the only cause worth noting is government action.”

Closure Range/Prescriptiveness:

- *Grammar A* is moderately prescriptive in a utilitarian sense (it prescribes vaccination as ethical/political duty).
- *Grammar B* is very prescriptive (basically “stop everything, investigate”) and highly negative.
- *Grammar C* is prescriptive in a rights-based sense (it prescribes legal protections, forbidding mandates). But it does not suggest a collective health course beyond personal choice.

So B is most interventionist (stop mandates), C is interventionist in policy (block mandates) but not in health terms, A is interventionist in biomedical policy (continue vaccination).

Closure Timing:

- Grammar A reached its closure relatively early (by 2021) as data rolled in (with adjustments but confidence remained high). It saw additional data (waning, Omicron) as refinements but not shaking its conclusion.
- Grammar B was most convinced as soon as it identified even rare adverse signals (mid-2021) and remained so. New evidence (waning, boosters) only solidified distrust. It likely “formed closure” in 2021–2022 when vaccines became widespread.
- Grammar C’s closure matured when mandates were first announced (mid-2021). Before that, it might have been more neutral or only theoretically individualist. Post-2021, every policy reaffirmed its core stance. Evidence since has generally been interpreted as either vindicating (e.g. any mandate reversal) or irrelevant (the debate is over coercion, not results).

In short: A was data-driven and updates continuously, B reached a negative verdict and saw data as confirmation, C hinged on policy events and remained consistent on principle.

Grammar Translation (Re-describing Closures):

- *How A would re-describe B’s closure:* A might say, “Grammar B’s conclusion is that vaccines harm more than they help.” In A’s terms: “From their perspective, all the morbidity prevented by vaccination is either ignored or negated by a presumed hidden harm. They effectively define vaccine 'truth' by adverse events, not by outcomes of disease prevention.” A might attempt to interpret B’s emphasis on profit as “explanatory noise” rather than a fundamental cause.

- *A re-describing C’s closure:* A would see C’s stance as “valuing freedom so much that even life-saving measures are suspect. They conclude vaccines must be personal choice because forcing

people even to save lives is tyrannical." In biomedical terms: "They consider the ethical veto overriding epidemiological benefit."

- *How B would re-describe A:* B would claim "Grammar A is naively trusting official sources, ignoring signs of harm. They see vaccines as heroic with blinders on." B might frame A's trust in trials as corruptible or incomplete. B would interpret A's mitigated view of adverse events as denial.

- *B re-describing C:* B might say "Grammar C is primarily concerned with its ideological value of freedom; they speak of choice as the ultimate good, essentially indifferent to whether vaccines actually work or not." B might accuse C of letting corporate/government agendas off the hook by focusing only on mandates.

- *C re-describing A:* C would interpret A as "putting science above liberty; they're fine with coercion because they see population data, not individuals." It might say A is paternalistic: "Grammar A essentially trusts big institutions to decide for you."

- *C re-describing B:* C could see B as "pointing out the hypocrisy and risk, almost agreeing that something is wrong with the vaccine campaigns – but B doesn't take the full step of opposing coercion, focusing instead on the product itself." C might find common ground (both distrust big Pharma), but say B fails to affirm choice. Conversely, C might regard B as a sort of subordinate voice: "B wants to ban shots, and C wants to ban mandates; both oppose authority."

Ontological Check:

- All three grammars nominally address "COVID-19 vaccine rollout and discourse." However, Grammar B's lens can verge on treating "the phenomenon" as fundamentally a corporate/government conspiracy rather than a public health effort. In effect, grammar B is almost analyzing a *different object* ("Big Pharma campaigns") under the same name. Grammar C similarly shifts ontology by turning it into an issue of rights rather than biology. So while all refer to "the vaccine situation," what counts as "the phenomenon" changes: A sees a biological intervention, B sees a socio-political manipulation, C sees a case of civil governance. They share the term but interpret different essence. This ontological divergence underscores that each grammar partly talks about a different phenomenon (science vs. profit vs. liberty) under the banner of "vaccine discourse."

Section 6: Counter-Signals, Leakage, and Stress Test

Consolidated Counter-Signals: Tensions noted during processing include:

- **Saved Lives Evidence:** The fact that vaccines *likely prevented millions of deaths*
- challenged Grammar B (since it expects a much larger perception of risk). Grammar B answered by asserting brainwashing or social desirability bias, highlighting the 10–20% dissenters as "waking up." Grammar C saw it as initial consent of population which they allowed (no complaint about that itself), but still insisted dissent must be allowed. Grammar A saw it as confirming strategy.

- **Absence of Catastrophe:** No grammar fully predicted or easily absorbed that large-scale rollouts did not cause obvious widescale harm beyond known rare events. For Grammar B, the

counter-signal was “where are the mass vaccine victims?” (It responded by saying they were hidden or future). For A, the initial reality (few excess hospitalizations beyond waves) confirmed its stance. For C, it’s irrelevant unless it ties to mandates.

Leakage (Blurring):
Some cross-over occurred. For example:

- Grammar B occasionally borrowed some direct-cause language: though typically conspiratorial, in processing waning it used A’s mechanistic insight (“vaccine antibodies wane”) to argue “see, it fails.”
- Grammar A sometimes had to acknowledge the individual angle (not strongly, but by saying “voluntary choice increases trust”).
- Grammar C, despite focusing on liberty, implicitly leaned on some data: it would admit if vaccines stopped a death in someone, that didn’t override freedom, but it still had to know outcomes at least to argue about necessity.

However, overall each largely stayed in its own register. They touched some of each other’s evidence (e.g. all three mention mandates in some way), but interpreted it within their own frame.

Grammar Stress Test: For each grammar, identify the single most threatening counter-signal, and whether it can absorb it:

- **Grammar A’s Threat:** The biggest challenge is Grammar B’s claim of widespread, unreported vaccine harm via VAERS and excess mortality. If Grammar A couldn’t explain these signals, its closure would weaken. Can it absorb them? Partially: it accounts for known rare events (safety signals) but denies broad-scale unobserved harm. It relies on scientific consensus and transparency of data to dismiss the “mass harm” claim. Since Grammar A considers its data sources reliable, it largely *excludes* the idea that millions died from vaccines (it would say that contradicts everything else). So Grammar A’s closure *depends on excluding* any uncontrolled theory of harm. If empirical evidence ever surfaced that vaccines caused e.g. thousands of unexplained deaths (beyond known patterns), Grammar A’s closure would fracture.

- **Grammar B’s Threat:** The most threatening signal is the evidence of net lives saved or hospitalizations prevented (like WHO’s modeling
-). This is the heart of the mainstream case. Can Grammar B absorb it? No, it must dismiss it as propaganda or error. Hence its closure inherently excludes accepting “vaccines saved millions”; it explains it away as math games. If somehow incontrovertible evidence of a huge positive benefit were presented (beyond modeling, say a natural experiment), Grammar B would likely reject the framing of the signal. Its closure only holds if those benefits are considered suspect.

- **Grammar C’s Threat:** The biggest threat would be if a vaccine mandate demonstrably eradicated a variant wave or prevented catastrophic collapse. For instance, if a country that mandated vaccines had zero Omicron deaths versus neighbors with high deaths, that would be at odds with its position. Grammar C would struggle to incorporate “mandates *saved lives*.” It might still say “but they overstepped,” or see that as one jurisdiction’s good luck. So Grammar C’s closure is conditional: it assumes mandates were not strictly necessary. If a strict correlation between mandates and saved lives were clearly shown, it would cause dissonance. It could try to “explain” such data by saying maybe voluntary uptake did it, or attributing success to other measures. But a clear mandate-to-benefit causal story would undermine its case. So Grammar C’s closure depends on the signal being either ambiguous or on its being interpreted apart from health outcomes.

In summary, each grammar can only maintain its closure by discounting or reinterpreting a key threatening signal: A discounts conspiracy claims; B discounts modeled benefits; C discounts mandate-success

correlations. If those counter-signals become empirically irresistible, each grammar's argument collapses (meaning its closure is *conditional* on those signals being handled away).

Section 7: Post-Analysis Reflexive Note

Looking back at Step 0, my home grammar (mainstream public health) indeed dominated my initial instincts. Grammar C (libertarian emphasis) was easiest to operate because I could rely on principles of consent easily. The *hardest* was Grammar B (safety-skeptic). I found it challenging to consistently foreground anecdotal or marginal signals over vast datasets; it felt unnatural to trust raw VAERS counts or to treat known benefits as "ignored." It forced me to set aside what I personally consider strong evidence.

I did discover some blind spots in myself: I had underplayed how much the framing of signals (like regulatory speed or profit figures) can sway conclusions. It was insightful to see how Grammar B and C sharply reinterpreted even the same data. For example, I now notice I might have underweighted how skepticism can legitimately question official narratives, or how autonomy arguments can stand independently of science.

If I were a member of a different home grammar, I'd probably structure the analysis differently from the start (e.g. an anti-vax researcher would call the biomedical view "big industry spin"). The final sections reflect those differing frames, but I personally had to be vigilant about not slipping in my default trust of science.

This exercise highlights how much conclusions are shaped by the interpretive architecture. Each grammar produced a distinct "story" from the same data, illustrating that evidence doesn't speak for itself; it must pass through a lens. I realized how easy it is to dismiss signals that don't fit one's framework (for example, I had initially wanted to dismiss VAERS totals as "mere reports," but saw how Grammar B hinges on them). The prismatic method forces us to recognize that "who says it, and how they process it," is as important as the fact itself.

Section 8: Methodological Observations

The prismatic method's structured steps were very helpful. Using precise parameters (primary variable, signal rules, etc.) ensured each grammar was coherently specified rather than a straw man. It guided me to articulate why *this* evidence leads to *that* conclusion, which is often implicit in debate. The insistence on counter-signals and contested signals during processing was particularly useful; it prevented simply stacking arguments in one direction. Noting ontological divergence was a good reminder that arguments sometimes don't just disagree—they're not even talking about the same "object" in the same way.

Difficulties arose in keeping the grammars balanced. Especially for Grammar B, there was a tension between fairly representing its internal logic and not outright endorsing misinformation. It required disciplined consistency. Also, some recommended parameters (like specifying "excluded signals") were a bit abstract; it took care to meaningfully say "this grammar excludes X." The formality of required parameters felt rigid at times, but it did ensure completeness.

One ambiguity was the line between Grammar B and C; in practice real discourse mixes "I don't want mandates" with "I'm worried about safety." The methodology prescribes separate grammars, but reality is more blurred. We handled it by strictly rule-based separation, but it's tricky. More guidance on dealing with overlapping grammars (e.g. if a stakeholder embodies multiple frames) would help.

Overall, I found it enlightening. One suggestion: provide illustrative examples of interpretive grammars for a simpler historical case (not a pandemic). Seeing a worked model helps guide the novel task. Also, maybe clarify expectations for grammar translation: do we need formal “metaphrases” or just descriptive re-telling? I chose descriptive.

In sum, the prismatic analysis forces rigor about the *process* of interpretation, not just the evidence. It highlighted that to understand disagreements, we must map the underlying logic. That metacognitive demand was challenging but ultimately productive. Suggestions for refinement: include explicit checks on how to treat uncertainty (“Bayesian revision” style) and more scaffolding on contested vs. omitted signals. But overall the method is very powerful for dissecting polarized debates.

. *Type:* Peer-reviewed RCT data. *Ambiguity:* Generally accepted in technical fields, but skeptics may question trial demographics or duration.

Clinical Trial Efficacy (Moderna): Phase III trial reported 94.1% efficacy (95% CI 89.3–96.8)
. *Type:* RCT. *Ambiguity:* Similar to above; in some debates, Moderna’s young demographic might be cited.

Clinical Trial Efficacy (Oxford/AstraZeneca): Pooled Phase III showed ~70.4% efficacy (95.8% CI 54.8–80.6)
. *Type:* RCT meta-analysis. *Ambiguity:* This result varied by dosing subgroup (one regimen ~62%, another ~90%). Critics often focus on lower bound (62%).

Clinical Trial Efficacy (Johnson & Johnson): Single-dose trial showed ~52.9% efficacy for moderate-to-severe disease (global) at ≥28 days
. *Type:* RCT. *Ambiguity:* The moderate-efficacy figure (52.9%) is much lower than mRNA vaccines, raising questions about vector-vaccine comparisons.

EUA / Authorization Processes: Timeline of regulatory approvals: e.g. FDA granted EUA to Pfizer (Dec 11, 2020) and Moderna (Dec 18, 2020)
. *Type:* Government regulatory documents. *Ambiguity:* Some see fast-track as evidence of urgency/success, others see it as “skipping usual checks.”

Real-World Effectiveness & Waning: Study from Sweden showed mRNA vaccines gave >90% protection initially, but Pfizer efficacy against infection waned near zero by 7 months (Moderna declined more slowly)
. *Type:* Epidemiological study (pre-Omicron). *Ambiguity:* Indicates waning immunity, but doesn’t fully account for boosters; skeptics may argue this means vaccines “don’t work long,” while others note initial protection vs severe disease remains moderate.

Variant-Specific Performance (Omicron): Hong Kong data (Jan–Jul 2022, Omicron BA.2 wave) showed 3 doses of BNT162b2 (Pfizer) gave ~48% effectiveness vs infection at 7 days, falling to ~26% by 100 days; 4 doses gave ~69%→35%. Inactivated vaccine (CoronaVac) gave lower VE
. *Type:* Observational VE study. *Ambiguity:* Highlights Omicron’s immune evasion and waning booster protection. Some view this as proof vaccines fail against variants; others note preserved (if reduced) protection.

Adverse Events (VAERS & Surveillance): VAERS (US passive system) collected reports of events post-vaccination; e.g. thousands of myocarditis and other adverse events have been reported. CDC emphasizes VAERS is for signal detection, not proof (a report “does not mean the vaccine caused the event”)
. *Type:* Passive surveillance data. *Ambiguity:* Highly contested. Some grammars treat VAERS counts as direct evidence of harm, others see them as unverified signals requiring further study.

Identified Safety Signals: CDC and FDA have confirmed rare vaccine-related syndromes: vaccine-induced thrombosis with thrombocytopenia (TTS) after adenovirus vaccines, and myocarditis after mRNA vaccines. For example, ACIP in Dec 2021 recommended mRNA over J&J after linking J&J to TTS
. *Type:* Agency safety reviews. *Ambiguity:* These officially acknowledged risks are small but concrete signals. Some interpret "rare but real"; others downplay them relative to COVID risk.

Excess Mortality (Global 2020–21): WHO (May 2022) modeled ~14.91 million global excess deaths in 2020–2021, versus ~5.42 million reported COVID deaths
. *Type:* Demographic modeling. *Ambiguity:* Indicates undercount of COVID impact; opponents sometimes twist this as "vaccine deaths," while mainstream sees most excess as uncounted COVID or indirect pandemic effects.

Excess Mortality (Post-Rollout): A 2024-25 study of 21 countries found that all had **continued excess mortality in 2022**, often non-COVID deaths likely due to healthcare disruptions or long-term pandemic effects
. *Type:* Vital stats analysis. *Ambiguity:* Suggests vaccines (and measures) saved lives long-term. Anti-vax voices might claim any excess means vaccines failed or harmed, but the study attributes most excess to indirect factors, not vaccines.

Pharmaceutical Company Revenues: Pfizer reported Comirnaty (COVID vaccine) sales of $36.8 billion in 2021
. *Type:* Corporate financials and press. *Ambiguity:* Shows huge profits. Public-health grammars see this as companies recouping R&D and scaling, whereas critical grammars see it as profiteering motives or conflict of interest.

Government Policies (Mandates/Passports): Many countries implemented vaccine passports or mandates. E.g., France (mid-2021) introduced a **"sanitary pass"** requiring vaccination or negative tests for most public venues
. *Type:* Policy records and news. *Ambiguity:* To some, this shows justified public health action boosting uptake; to others, it looks like coercion or rights violations.

WHO/National Messaging Evolution: WHO and national bodies regularly updated guidance (e.g. boosters, masking). For example, CDC reduced isolation guidelines over time, and WHO shifted booster emphasis. *Type:* Official guidance. *Ambiguity:* Frequent changes (changing mask guidelines, etc.) fuel distrust in some audiences. (For instance, the public debate over ACIP's changes to childhood vaccine schedules has been litigated
, indicating trust tensions.)

Social Media Content Policies: Platforms like Facebook/Meta and Twitter implemented strict COVID/vaccine misinformation rules by 2021. Meta announced in Feb 2021 it would remove COVID vaccine "claims such as…'Vaccines are toxic…or cause autism'"
. *Type:* Corporate policy announcements. *Ambiguity:* Many see this as responsible misinformation control; others accuse "censorship" and suppression of dissenting signals.

Public Opinion Polls (Trust and Attitudes): Surveys consistently found strong public support for vaccines (e.g. an Impact/Research!America poll in Jan 2022 found ~90% of Americans thought policymakers must ensure vaccine access and 89% said vaccines are essential for public health
, with sharp partisan divides. *Type:* Polling. *Ambiguity:* These data indicate broad baseline support but eroding trust; some grammars will focus on the high acceptance rates, others on the decline in confidence.

Vaccination Rate Disparities: Data show unequal uptake. A systematic review (2025) found Black and mixed-race populations in US/UK were significantly less likely to be vaccinated than Whites (Blacks ~13% less likely)

(reducing COVID fatalities by ~63% in that period). This evidence of benefit would counter any closure that vaccines were net harmful.
High Public Support: Polls showing ~90% support for vaccine access and 93% support for vaccination requirements
serve as counterpoint to narratives emphasizing "small anti-vax minorities."
Low General Adverse Impact: By late 2024, widespread surveillance found no new large-scale harms. For instance, CDC data (Slide deck) emphasizes only specific rare events (myocarditis, TTS) as linked; most other reported events had no confirmed link
). *According to vaccine-skeptic grammar,* the *volume* of VAERS reports (e.g. "tens of thousands of adverse events, many serious") counts as evidence of harm.
Excess Mortality: *Public-health grammar* treats excess deaths as mostly from uncounted COVID or pandemic effects (WHO: 14.9M excess in 2020–21, implying large undercount of COVID deaths
, it might say "Well, if protection fades so fast, it's even harder to justify mandates." Overall, *vaccine efficacy* is acknowledged (most libertarians aren't anti-science per se), but it is not the priority variable.

EUA/Authorization: The *fact* of fast EUAs supports Grammar C's suspicion of authority: "Authorities rushed this through – see, they didn't have full data." If it hears Pfizer's long-term follow-up gave 91%
, it might note "They changed the number after full data – why change?" It uses regulatory signals to question legitimacy. Full approvals are seen more as political milestones than safety endorsements.

Waning/Variants: These signals (weakening protection) are used to argue that "even science admits vaccines aren't a perfect fix, so coercive measures were unjustified." It might say "If vaccines don't stop Omicron infection much
, what was the point of forcing them on people?" The metric here is usefulness to justify mandates: anything undermining the clarity of the "vaccine=solution" narrative strengthens the case for individual choice.

Safety/Adverse Events: Grammar C does care about safety, but primarily as fodder against authority: e.g. "Regulators admitted rare cases of myocarditis
might be cited as background noise that proves regulators don't know all consequences.

Excess Mortality: This grammar might say "Excess deaths (WHO 14.9M)
even after mass vaccination; so vaccine mandates didn't solve it." It's more concerned with whether policies led to "any net difference." If any negative statistic comes from a time of mandates, that's seen as failure of authority.

Pharma Profits: Grammar C would definitely use revenue signals as evidence of vested interests, especially regarding mandates. E.g. "Pfizer made $37B
– politically they're a power player. Government colluding with them would violate the public interest (and our right to autonomy)." But unlike Grammar B, the focus is on the *act of mandating for profit* rather than on the intrinsic harm to individuals. Profit is a *motive* behind the policy in Grammar C's eyes.

Mandates/Passports: By far the central signal for this grammar. France's pass system
is cited as *evidence of government overreach*: "France co-opted society into forcing shots!" It doesn't debate the health effect; it insists "even if it protected some people, it violates freedom." The grammar views *any* broad compulsion as negative. If forced-vaccination policies are revoked later (as some were in 2022), that's taken as vindication. If policies persist, they justify more resistance. The grammar's chain: mandates themselves ➔ erosion of civil liberties and trust. The presence of mandates is itself a "signal" of wrongdoing.

Public Messaging: This grammar scrutinizes health rhetoric: e.g. **advertising "safe and effective"** vs. reality. It notes if CDC says "vaccines are safe" (which it does repeatedly) and contrasts with any personal rights discourse: "They keep pushing 'safe and effective' but still fired our citizens for refusing." It seizes public debates (like the CDC's policy shift on boosters) to argue "the goalposts keep moving, so why trust

them?” However, unlike Grammar B, Grammar C is not as interested in safety data per se, but in the consistency of authority. Public confusion is read as a failure of communication, supporting the idea that “government can’t be trusted with mandates.”

Social Media Moderation: This grammar strongly views [78†L309-L317] (Facebook’s plan to remove claims including “vaccines cause autism”) as proof of censorship. It might cite that meta bullet list of claims banned
to argue “They even banned *stating adverse effects*, so of course people don’t trust them.” The act of moderation is itself a symbol of intrusion. The grammar may even embrace some conspiracy allegations: e.g. “If they’re censoring talk of vaccines, maybe it’s because they know damage can’t be hidden if people talk!” Censorship signs strengthen the conclusion that authorities overstepped.

Public Opinion: Polls showing support for vaccine access
is likewise seen as justification to leave decisions to individuals.

Disparities: This grammar might see the lower uptake in minority/rural groups as a sign that these groups rightly distrust official campaigns. E.g. rural vs urban
, while B sees them as confirmation of widespread harm, and C treats them as evidence that authorities cannot be sure of safety, bolstering its emphasis on personal vigilance.
Excess Mortality: Grammar A attributes excess deaths to indirect pandemic effects
was a counterpoint especially for Grammar B. It was downplayed by B (considered a model projection), but Grammar A used it to bolster its view. Grammar C would say “even if it saved lives, compulsion is still wrong.” So this signal created a cross-grammar mismatch.

Majority Support in Polls: Over 80–90% support for vaccine access